

Chapter 1

Theorem 1. let $\rho : G \rightarrow GL(V)$ be a representation of G in V , and let $W \subset V$ be a vector subspace of V stable under G . Then there exists a complement W^0 of W in V which is stable under G .

Proof. Let W^0 be a complement of W in V , and let $p : V \rightarrow V$ be the projection of V onto W corresponding to the decomposition $V = W \oplus W^0$. We define a new projection $p^0 : V \rightarrow V$:

$$p^0 v := \frac{1}{|G|} \sum_{s \in G} \rho_s p \rho_s^{-1} v.$$

Notice that p maps to W and W is stable under G , so $\rho_s p$ sends elements to W . If $w \in W$, then $\rho_s^{-1} w \in W$. Hence,

$$p^0 w = \frac{1}{|G|} \sum_{s \in G} \rho_s p \rho_s^{-1} w = \frac{1}{|G|} \sum_{s \in G} \rho_s \rho_s^{-1} w = w.$$

So, p^0 is a projection of V to W .

Let $x \in W^0$. Then, $p^0 \rho_s x = \frac{1}{|G|} \sum_{s \in G} \rho_s p \rho_s^{-1} \rho_s x = \frac{1}{|G|} \sum_{s \in G} \rho_s p x = 0$. Since $p^0(\rho_s x) = 0$ for any $x \in W^0$, we have $\rho_s x \in W^0$. Hence, W^0 is stable under G . $\square$